ECE253/CSE208 Introduction to Information Theory

Lecture 5: Data Processing Inequality

Dr. Yu Zhang
ECE Department
University of California, Santa Cruz

- Chap 2 of *Elements of Information Theory (2nd Edition)* by Cover–Thomas
- Chap 2–4 of *Convex Optimization* by Boyd–Vandenberghe

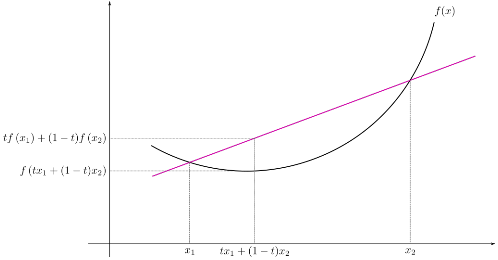
Convex Functions

Definition (Convexity of functions)

- A function $f(x)$ is *convex* over an interval (a, b) iff

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

holds for any $x_1, x_2 \in (a, b)$ and $\lambda \in [0, 1]$.



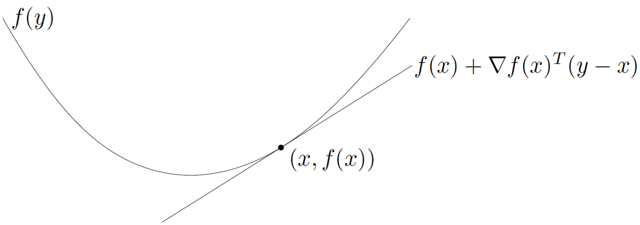
Convex Functions

Definition (Convexity of functions)

- A function $f(\cdot)$ that is *differentiable everywhere* in (a, b) is *convex* iff

$$f(y) \geq f(x) + f'(x)(y - x)$$

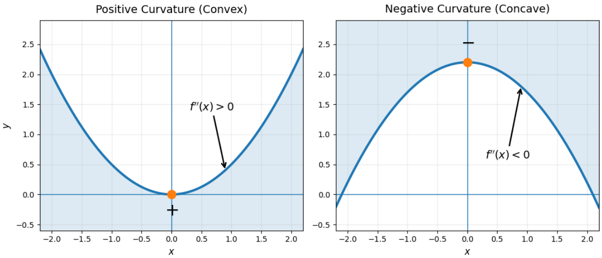
for any $x, y \in (a, b)$.



Convex Functions

Definition (Convexity of functions)

- A *twice differentiable* function $f(x)$ over (a, b) is *convex* iff $f''(x) \geq 0 \, \forall x \in (a, b)$.



Examples. Convex functions: $ax + b, |x|, x^2, x^4, e^{\pm x}, x \log x$.

- If $f(x)$ is *convex*, then $-f(x)$ is *concave*.
- Affine functions are both convex and concave.

Convexity in High-dimensional Spaces

Definition (Convex function)

A function $f(\mathbf{x}) : \mathbb{R}^n \mapsto \mathbb{R}$ is *convex* iff for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we have one of the following:

- 0th-order condition: $f(\lambda \mathbf{x} + (1-\lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1-\lambda) f(\mathbf{y})$ for any $\lambda \in [0, 1]$.
- 1st-order condition: $f(\mathbf{y}) \geq f(\mathbf{x}) + \triangledown f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$
- 2nd-order condition: $\mathbf{H}_f := \triangledown^2 f(\mathbf{x}) \succeq \mathbf{0}$; i.e., the *Hessian* matrix is positive semi-definite (all eigenvalues are nonnegative), where $\mathbf{H}_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

- Strictly convex: if strict inequality always holds when $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.
- Strongly convex: $\mathbf{H}_f \succeq a\mathbf{I}$ for some constant $a > 0$ (the Hessian is positive definite).
- Geometrically, strict/strong convexity implies that the function has no flat part and curves upward everywhere $\implies$ unique minimizer.

Convex Sets

Definition (Convex sets)

A set $S \subseteq \mathbb{R}^n$ is convex iff $\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 \in S, \forall\ \mathbf{x}_1, \mathbf{x}_2 \in S, \lambda \in [0,1]$.

Geometrically, a convex set contains line segment between any two points in the set.
Convex sets are solid body without holes and curve outward.

examples (one convex, two nonconvex sets)

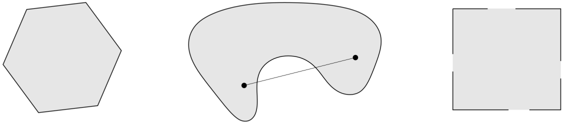


Figure: Convex and nonconvex sets (source: Stephen Boyd, Stanford).

Epigraph and Sublevel Set

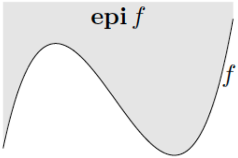
- α**-sublevel set** of $f : \mathbb{R}^n \to \mathbb{R}$:

$$C_\alpha = \{\mathbf{x} \in \mathrm{dom} f : f(\mathbf{x}) \leq \alpha\}$$

  sublevel sets of convex functions are convex (converse is false)

- **epigraph** of $f : \mathbb{R}^n \to \mathbb{R}$:

$$\mathrm{epi} f = \{(\mathbf{x}, t) \in \mathbb{R}^{n+1} : \mathbf{x} \in \mathrm{dom} f,\ f(\mathbf{x}) \leq t\}$$



f is a convex function $\iff$ $\mathrm{epi} f$ is a convex set

Convex Optimization Problem

Convex optimization problem in standard form:

$$\begin{aligned} \min_{\mathbf{x}} \quad & f_0(\mathbf{x}) \\ \text{s.to} \quad & f_i(\mathbf{x}) \le 0, \quad i = 1, \ldots, m \\ & \mathbf{a}_i^\top \mathbf{x} = b_i, \quad i = 1, \ldots, p \end{aligned}$$

- $f_0, f_1, \ldots, f_m$ are convex
- equality constraints are affine (alternatively $\mathbf{A}\mathbf{x} = \mathbf{b}$)

Important properties:

1. For convex problems, any local solution is also global.
2. If $f_0(\mathbf{x})$ is strictly convex, the minimizer is unique.
3. The optimal set X_{opt} is convex.

Jensen's Inequality

Lemma (Jenson's Inequality)

If X is a random variable and $f(\cdot)$ is a convex function, then

$$\boxed{E(f(X)) \geq f(E(X))}$$

Moreover, if $f(X)$ is strictly convex, equality implies $X = \mathrm{E}(X)$ with probability 1.

Proof of Jensen's Inequality

Proof by induction:

1) For a *two-point distribution* $X \in \{x_1, x_2\}$, the convexity of $f(\cdot) \implies$

$$p_1 f(x_1) + p_2 f(x_2) \geq f(p_1 x_1 + p_2 x_2).$$

2) Assume Jensen's inequality holds for a $(k-1)$-mass point distribution. Define $p_i' = \frac{p_i}{1-p_k}$ for all $i = 1, 2, \ldots, k-1$:

$$\begin{aligned}
\sum_{i=1}^{k} p_i f(x_i) &= p_k f(x_k) + (1-p_k) \sum_{i=1}^{k-1} p_i' f(x_i) \\
&\geq p_k f(x_k) + (1-p_k) f\left(\sum_{i=1}^{k-1} p_i' x_i\right) \text{ [from induction]} \\
&\geq f\left(p_k x_k + (1-p_k) \sum_{i=1}^{k-1} p_i' x_i\right) = f\left(\sum_{i=1}^{k} p_i x_i\right) \text{ [due to convexity of f}(\cdot)\text{]}
\end{aligned}$$

Gibbs' Inequality (Information Inequality)

Theorem (Gibbs' Inequality)

Let $p(x)$ and $q(x)$ be two probability mass functions. Then, $D_{\mathrm{KL}}(p\|q) \geq 0$ with equality iff $p(x) = q(x)$ for all x.

Proof: $-D_{\mathrm{KL}}(p\|q) = \sum_{x\in\mathcal{X}} p(x)\log\frac{q(x)}{p(x)} \leq \log\left(\sum_{x\in\mathcal{X}} p(x)\frac{q(x)}{p(x)}\right) = \log\left(\sum_{x\in\mathcal{X}} q(x)\right) = 0.$

Corollary (Nonnegativity of mutual information)

For any two random variables X and Y, we have $I(X;Y) \geq 0$ with equality iff X and Y are independent.

Information Cannot Hurt

Corollary (Conditional mutual Information)

$$I(X;Y|Z) \geq 0$$

with equality iff X *and* Y *are conditionally independent given* Z*, which is denoted as* $(X \perp Y) \mid Z$*.*

Theorem (Conditioning reduces entropy)

$H(X|Y) \leq H(X)$ *with equality iff* $X \perp Y$*.*

Intuitively, knowing Y can only reduce the uncertainty in X. Note that this is true only on the average (expectation) sense. That is, $H(X|Y=y) > H(X)$ can happen.

Maximum Entropy Distribution

Theorem (Uniform distribution has the maximum entropy)

$$H(X) \leq \log |\mathcal{X}|$$

where $|\mathcal{X}|$ is the cardinality of the set $\mathcal{X}$ (i.e., the number of elements in the set) with equality iff X has a uniform distribution over $\mathcal{X}$.

Proof: Let $u(x) = \frac{1}{|\mathcal{X}|}$ be the uniform PMF, and $p(x)$ be the PMF of X over $\mathcal{X}$, respectively. Then, $D_{\mathrm{KL}}(p\|u) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{u(x)} = \log |\mathcal{X}| - H(X) \geq 0$.

Entropy Union Bound

Theorem (Independence bound on entropy)

Let RVs $\{X_i\}_{i=1}^n$ be drawn from $p(x_1, \ldots, x_n)$. Then, $H(X_1, \ldots, X_n) \leq \sum_{i=1}^n H(X_i)$ with equality iff $\{X_i\}_{i=1}^n$ are independent.

Proof: By chain rule: $H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \ldots, X_1) \leq \sum_{i=1}^n H(X_i)$.

Log-sum Inequality

Lemma (Log-sum inequality)

For nonnegative numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$, we have

$$\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} \geq \left(\sum_{i=1}^{n} a_i \right) \log \left(\frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i} \right)$$

with equality iff $\frac{a_i}{b_i} =$ constant.

Proof: using convexity of $f(t) = t \log t$ ($f''(t) > 0$); see page 31 of Cover's book.

Example: $a_1 \log \frac{a_1}{b_1} + a_2 \log \frac{a_2}{b_2} \geq (a_1 + a_2) \log \left(\frac{a_1 + a_2}{b_1 + b_2} \right)$.

Concavity of Entropy

Theorem (Concavity of entropy)

$H(\mathbf{p})$ is a concave function of $\mathbf{p}$, where $\mathbf{p} := [p_1, \dots, p_k]$ denotes an arbitrary k-point discrete probability mass function.

Proof 1: Let $X_i \sim \mathbf{p}_i (i = 1, 2)$ be two RVs defined on the same set. For any $\lambda \in [0, 1]$, define

$$\theta = \begin{cases} 1, & \text{with prob } \lambda \\ 2, & \text{with prob } 1-\lambda. \end{cases}$$

Hence, the new RV $X_\theta \sim \lambda \mathbf{p}_1 + (1-\lambda)\mathbf{p}_2$.
$\because H(X_\theta) \geq H(X_\theta|\theta) \implies H(\lambda \mathbf{p}_1 + (1-\lambda)\mathbf{p}_2) \geq \lambda H(\mathbf{p}_1) + (1-\lambda) H(\mathbf{p}_2)$.

Alternative Proofs of Entropy Concavity

Proof 2: $H(\mathbf{p}) = -\sum_i p_i \log(p_i)$. Note that each term in the sum is convex and only depends on one p_i. Hence, $H(\mathbf{p})$ is concave.

Proof 3: $H(\mathbf{p}) = \log|\mathcal{X}| - D_{\mathrm{KL}}(\mathbf{p}||\mathbf{u})$, where $\mathbf{u}$ is the uniform distribution. Since $D_{\mathrm{KL}}(\mathbf{p}||\mathbf{u})$ is convex in $\mathbf{p}$, $H(\mathbf{p})$ is concave.

Convexity of KL Divergence

Theorem (Convexity of relative entropy)

$D_{\mathrm{KL}}(p||q)$ is convex in the pair (p, q). That is, if (p_1, q_1) and (p_2, q_2) are two pairs of PMFs. Then, $D_{\mathrm{KL}}(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) \leq \lambda D_{\mathrm{KL}}(p_1 \| q_1) + (1-\lambda) D_{\mathrm{KL}}(p_2 \| q_2)$ holds for any $\lambda \in [0, 1]$.

Proof: By log-sum inequality, we have

$$[\lambda p_1(x) + (1-\lambda)p_2(x)] \log \left(\frac{\lambda p_1(x) + (1-\lambda)p_2(x)}{\lambda q_1(x) + (1-\lambda)q_2(x)} \right)$$
$$\leq \lambda p_1(x) \log \frac{p_1(x)}{q_1(x)} + (1-\lambda)p_2(x) \log \frac{p_2(x)}{q_2(x)}$$

Summing this over all $x \in \mathcal{X}$ completes the proof.

Convexity of Mutual Information (1/2)

Theorem (Convexity of mutual information)

Let $(X, Y) \sim p(x, y) = p(x)p(y|x)$. Then $I(X; Y)$ is a concave function of $p(x)$ for fixed $p(y|x)$, and a convex function of $p(y|x)$ for fixed $p(x)$.

Proof.

Part I (concavity in $p(x)$ for fixed $p(y|x)$). Since

$$I(X; Y) = H(Y) - H(Y|X) = H(Y) - \sum_x p(x)H(Y|X = x),$$

- $p(y) = \sum_x p(x)p(y|x)$ is linear in $p(x)$ for fixed $p(y|x)$.
- $H(Y)$ is concave in $p(y)$; since $p(y)$ is linear in $p(x)$, $H(Y)$ is concave in $p(x)$.
- The second term $\sum_x p(x)H(Y|X = x)$ is linear in $p(x)$.

Hence, $I(X; Y)$ is concave in $p(x)$.

Convexity of Mutual Information (2/2)

Part II (convexity in $p(y|x)$ for fixed $p(x)$). For fixed $p(x)$, define $W(y|x) = p(y|x)$. Then

$$I(X;Y) = D_{\mathrm{KL}}\big(p(x)W(y|x) \,\|\, p(x)p(y)\big), \qquad p(y) = \sum_x p(x)W(y|x).$$

Let $W_\lambda = \lambda W_1 + (1-\lambda)W_2$ with $0 \le \lambda \le 1$. Then both

$$p^{(\lambda)}_{XY}(x,y) = p(x)W_\lambda(y|x) \quad \text{and} \quad p_X p^{(\lambda)}_Y(x,y) = p(x)\sum_{x'} p(x')W_\lambda(y|x')$$

are linear in W_λ. By joint convexity of KL divergence,

$$D_{\mathrm{KL}}\big(p^{(\lambda)}_{XY} \,\|\, p_X p^{(\lambda)}_Y\big) \le \lambda D_{\mathrm{KL}}\big(p^{(1)}_{XY} \,\|\, p_X p^{(1)}_Y\big) + (1-\lambda) D_{\mathrm{KL}}\big(p^{(2)}_{XY} \,\|\, p_X p^{(2)}_Y\big),$$

which implies $I(X;Y)$ is convex in $p(y|x)$ for fixed $p(x)$. ■

Markov Chain

Definition

Random variables X, Y, Z are said to form a Markov chain in the $X \to Y \to Z$ if the joint PMF can be written as $p(x, y, z) = p(x)p(y|x)p(z|y)$.

Note: $p(x, y, z) = p(x)p(y, z|x) = p(x)p(y|x)p(z|y, x)$.

Corollary

$X \to Y \to Z$ iff $(X \perp Z) \mid Y$. Markovity implies conditional independence since

$$p(x, z|y) = \frac{p(x, y, z)}{p(y)} = \frac{p(x, y)p(z|y)}{p(y)} = p(x|y)p(z|y).$$

Corollary ($X \leftrightarrow Y \leftrightarrow Z$)

$$X \to Y \to Z \Leftrightarrow Z \to Y \to X.$$

Data Processing Inequality (DPI)

Theorem (DPI)

If $X \to Y \to Z$ then

$$I(X;Y) \geq I(X;Z)$$

with equality iff $I(X;Y|Z) = 0$ (i.e., $X \to Z \to Y$).

Proof: By the chain rule, we expand mutual information in two different ways:

$$I(X;Y,Z) = I(X;Y) + I(X;Z|Y) = I(X;Z) + I(X;Y|Z) \qquad (*)$$

$$\because (X \perp Z) \mid Y \implies I(X;Z|Y) = 0 \implies I(X;Y) \geq I(X;Z).$$

Corollary of DPI (1/2)

Corollary

1. *If $X \to Y \to Z$, then $H(X|Z) \geq H(X|Y)$.*
2. *$I(X;Y) \geq I(X;g(Y))$ for any function $g(\cdot)$; i.e., post-processing decreases the info.*
3. *If $X_1 \to X_2 \to \cdots \to X_n$, for any i, j, k, l such that $1 \leq i \leq j \leq k \leq l \leq n$, we have $I(X_i;X_l) \leq I(X_j;X_k)$.*

Proof:

1. $H(X) - H(X|Z) = \boxed{I(X;Z) \leq I(X;Y)} = H(X) - H(X|Y)$.

Corollary of DPI (2/2)

Corollary

If $X \to Y \to Z$, then $I(X;Y|Z) \leq I(X;Y)$; proof: see equation $(*)$ on page 20.

Generally, it is possible to have $I(X;Y|Z) > I(X;Y)$; see the following example when X, Y, Z do not form a Markov chain.

Example

Consider $Z = X + Y$ for two i.i.d. $X, Y \sim \mathrm{Bern}(0.5)$, Find $I(X;Y|Z)$.

$$\begin{aligned} I(X;Y|Z) &= H(X|Z) - H(X|Y,Z) \\ &= \Pr(Z=0)H(X|Z=0) + \Pr(Z=2)H(X|Z=2) + \Pr(Z=1)H(X|Z=1) \\ &= [\Pr(\mathrm{X}=1,\mathrm{Y}=0) + \Pr(\mathrm{Y}=1,\mathrm{X}=0)] \times H(\mathrm{X}|\mathrm{Z}=1) \\ &= 2 \times \frac{1}{4} \times H(1/2,1/2) = 0.5 > I(X;Y) = 0 \end{aligned}$$

Statistics: Function of Samples

- A statistic is any quantity computed from values in a sample which is considered for a statistical purpose; i.e., statistic is a function of a sample; e.g., sample mean and sample variance. We estimate things all the time. Estimation help us understand the behavior of a large population with a small sample.
- Example 1: We want to know the average height of students in a school district with a population of 10,000. We take a sample of 100 students, measure them and find that the mean height, which is the sample mean, the statistic (estimator).
- Example 2: We survey the people in a neighborhood about their preference of candidate. Using this limited amount of data, we can estimate the likeliness of a candidate to win an election in a larger area.
- How do we know if every bit of information in data has been explored to estimate the parameter.

Sufficient Statistics

A statistic is sufficient for a parameter if it captures all the information in the sample about that parameter. In other words, once you know this statistic, the rest of the data tells us nothing more about the parameter.

Given a family of distributions $\{f_\theta(x)\}$, let X be a sample drawn from a distribution in this family, and $T(X)$ be any statistic. Hence, $\theta \to X \to T(X) \Rightarrow I(\theta; X) \geq I(\theta; T(X))$.

Definition (Sufficient Statistic)

A function $T(X)$ is a sufficient statistic relative to the family $\{f_\theta(x)\}$ if X is independent of θ given $T(X)$ for any distribution on θ ($\theta \perp X \mid T(X)$); i.e., $\theta \to T(X) \to X$ forms a Markov chain, and

$$\boxed{I(\theta; X) = I(\theta; T(X)).}$$

Alternative Definition of Sufficient Statistic

Definition (Sufficient Statistic, SS)

Let X be a random sample from a distribution with parameter θ. $T(X)$ is a sufficient statistic for θ if the conditional probability $\Pr(X \mid T(X) = k)$ does not depend on θ.

From Bayesian's view, we also have $\Pr(\theta \mid X, T(X) = k) = \Pr(\theta \mid T(X) = k)$

Implications of Sufficient Statistic

Implication

1. Given a set X of i.i.d. data conditioned on an unknown parameter θ, a sufficient statistic is a function $T(X)$ whose value contains all the information needed to compute any estimate of θ.
2. A statistics is sufficient for θ if it contains all information in X about θ: Once we know $T(X)$, the remaining randomness in X does not depend on θ.

Example 1 of Sufficient Statistic (1/2)

Example

Given i.i.d. $X_1, \ldots, X_n \sim \mathrm{Bern}(\theta)$. Let $X := (X_1, \ldots, X_n)$. A sufficient statistic for θ is

$$T(X) = \sum_{i=1}^{n} X_i.$$

Proof. We show that the conditional PMF of X given $T(X)$ does not depend on θ. For any realization $x = (x_1, \ldots, x_n) \in \{0,1\}^n$,

$$\Pr(X = x) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^{n} x_i}(1-\theta)^{n-\sum_{i=1}^{n} x_i}.$$

Also, $T(X) \sim \mathrm{Bin}(n, \theta)$, so $\Pr(T(X) = k) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$.

Example 1 of Sufficient Statistic (2/2)

Hence, for any $k \in \{0, 1, \ldots, n\}$,

$$\Pr(X = x \mid T(X) = k) = \begin{cases} 0, & \text{if } \sum_{i=1}^{n} x_i \neq k, \\ \binom{n}{k}^{-1}, & \text{if } \sum_{i=1}^{n} x_i = k, \end{cases}$$

which is independent of θ. Therefore, $T(X)$ is sufficient for θ.

Factorization Theorem

Theorem (Fisher–Neyman factorization theorem)

A statistic $T(X)$ is sufficient for θ iff functions g and h can be found such that the probability density or mass function can be factorized as

$$f_\theta(\mathbf{x}) = h(\mathbf{x})g_\theta(T(\mathbf{x})),$$

where $h(\cdot)$ does not depend on θ while $g(\cdot)$ which does depend on θ, depends on $\mathbf{x}$ only through $T(\mathbf{x})$.

Example 2 of Sufficient Statistic

Example

Example 2. If $X_i \sim \mathrm{Unif}(\theta, \theta+1)$ i.i.d., then a sufficient statistic for θ is

$$T(X) = \Big(\min_{1 \le i \le n} X_i,\ \max_{1 \le i \le n} X_i \Big).$$

Proof. For $x = (x_1, \ldots, x_n)$, the joint density is

$$f_\theta(x) = \prod_{i=1}^{n} \mathbf{1}\{\theta \le x_i \le \theta + 1\} = \mathbf{1}\big\{\theta \le x_{(1)} \text{ and } x_{(n)} \le \theta + 1\big\},$$

where $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. Equivalently,

$$f_\theta(x) = \mathbf{1}\big\{x_{(n)} - 1 \le \theta \le x_{(1)}\big\} = g_\theta\big(x_{(1)}, x_{(n)}\big)\, h(x),$$

with $g_\theta(u, v) = \mathbf{1}\{v - 1 \le \theta \le u\}$ and $h(x) \equiv 1$. Thus $f_\theta(x)$ factors through $T(X) = (X_{(1)}, X_{(n)})$, so $T(X)$ is sufficient for θ.

More Examples

- Bernoulli distribution:

$$f_\theta(\mathbf{x}) = \theta^{\sum x_i}(1-\theta)^{(n-\sum x_i)} = \underbrace{\theta^{T(\mathbf{x})}(1-\theta)^{(n-T(\mathbf{x}))}}_{g_\theta(T(\mathbf{x}))} \times \underbrace{1}_{h(\mathbf{x})}$$

- $\mathrm{Unif}(\theta, \theta+1)$:

$$f_\theta(\mathbf{x}) = \prod_{i=1}^{n} \mathbb{1}_{\{\theta \le X_i \le \theta+1\}} = \underbrace{\mathbb{1}_{\{\theta \le \min\{X_i\} \le \max\{X_i\} \le \theta+1\}}}_{g_\theta(T(\mathbf{x}))} \times \underbrace{1}_{h(\mathbf{x})}$$

Minimal Sufficient Statistic (1/2)

- A statistic $T(X)$ is a minimal sufficient statistic if it is a function of every other sufficient statistic $U(X)$. In terms of DPI, $\theta \to T(X) \to U(X) \to X$.

- Minimal sufficient statistic is a statistic that has smallest mutual information with X while having largest mutual information with θ. That is,

$$T(X) \in \arg\min_{U(X)} I(X; U(X)) \text{ s.t. } I(\theta; U(X)) = \max_{T'(X)} I(\theta; T'(X))$$

- For every two sample points x and y, $\frac{f_\theta(x)}{f_\theta(y)}$ is independent of $\theta \iff T(x) = T(y)$. Then, $T(X)$ is a minimal sufficient statistic for θ.

Minimal Sufficient Statistic (2/2)

- Minimal sufficient statistic is not unique. Any one-to-one function of a minimal sufficient statistic is also a minimal sufficient statistic.
- Minimal sufficient statistic may not even exist.
- Minimal sufficient statistic most efficiently captures (i.e., maximally compresses) the information about θ in the sample; see more discussions here[1].

[1] https://bookdown.org/jkang37/stat205b-notes/lecture04.html

Fano's Inequality (1/2)

Consider a Markov chain $X \to Y \to \hat{X}$. In the context of communication:

- Send X through a noisy channel that possibly yields the received symbol $Y \neq X$.
- Recover X by post-processing Y to get an estimate $\hat{X} = g(Y)$ for some function g.
- Define the probability of error as $P_e := \Pr(\hat{X} \neq X)$.

Fano's Inequality (2/2)

Fano's inequality: We may estimate X with small P_e when $H(X|Y)$ is small.

For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, define $P_e = \Pr(X \neq \hat{X})$, we have

$$H\left(P_e\right) + P_e \log|\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y) \implies P_e \geq \frac{H(X|Y) - 1}{\log|\mathcal{X}|}.$$

If $\hat{X} \in \mathcal{X}$, we then have a slightly better upper bound for $H(X|Y)$:

$$H\left(P_e\right) + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)^{a}.$$

[a] $H(X|Y)$ was called the 'equivocation', which corresponds to effect of channel noise; i.e., the uncertainty in the channel input from the receiver's perspective. Hence, the inequality relates the equivocation (as an information measure) to the probability of error (as the traditional measure).

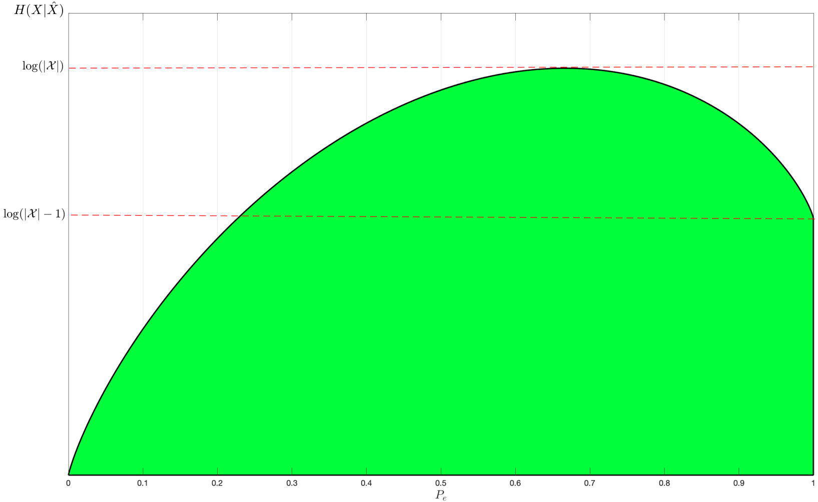
Implications of Fano's Inequality

$$\boxed{H(X|\hat{X}) \leq H\left(P_e\right) + P_e \log(|\mathcal{X}|-1) \leq H\left(P_e\right) + P_e \log|\mathcal{X}|}$$

Intuition and implication:

- $H(X|\hat{X})$ is how much uncertainty left in X after we know its estimate $\hat{X}$. Intuitively, if P_e is small, then $H(X|\hat{X})$ should also be small.
- Coding view: $H(X|\hat{X})$ quantify how many bits need for measuring X from $\hat{X}$. Fano' ineq gives us an upper bound of it. First, use $H(P_e)$ bits to show if $\hat{X} = X$. If yes, no more bits needed. If no (with probability P_e), we must use $\log|\mathcal{X}|$ bits to describe X in the worst case. Furthermore, if $\hat{X} \in \mathcal{X}$, then X can be any one of the other $|\mathcal{X}|-1$ symbols in the alphabet.
- Fano's ineq establishes the fundamental limits of data compression and transmission. It can be used to characterize when a perfect reconstruction of sent code is not possible, i.e. P_e is bounded away from zero. This is useful to prove converse results for coding theorems.

Permissible Region of Fano's Inequality



Fano's Inequality Proof

Proof: Define $E = \mathbb{1}_{\{\hat{X} \neq X\}}$. By chain rule, we have

$$\begin{aligned}
H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\
&= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log |\mathcal{X}|} \\
&\leq H(P_e) + \Pr(E=0)\,\underbrace{H(X|\hat{X}, E=0)}_{=0} + \Pr(E=1) H(X|\hat{X}, E=1) \\
&\leq H(P_e) + P_e \log |\mathcal{X}|
\end{aligned}$$

$\implies H(X|Y) \leq H(X|\hat{X}) \leq H\left(P_e\right) + P_e \log |\mathcal{X}|$.
If $X, \hat{X} \in \mathcal{X} \implies H(X|\hat{X}, E=1) \leq P_e \log(|\mathcal{X}| - 1)$.

Fano's Inequality Corollary

Corollary

For any two RVs X and Y, let $p = \Pr(X \neq Y)$. We have $H(p) + p \log|\mathcal{X}| \geq H(X|Y)$.

Proof: Let $\hat{X} = Y$ in Fano's inequality.

Fano's Inequality is Sharp

Example

Let $X \in \{1, 2, \ldots, m\}$ and $p_1 \geq p_2 \geq \cdots \geq p_m$. Then the best guess of X is $\hat{X} = 1$ and the resulting probability of error is $P_e = 1 - p_1$. Fano's inequality becomes

$$H(P_e) + P_e \log(m-1) \geq H(X).$$

The PMF $(p_1, p_2, \ldots, p_m) = \left(1 - P_e, \frac{P_e}{m-1}, \ldots, \frac{P_e}{m-1}\right)$ achieves the lower bound with equality. To see this,

$$\begin{aligned} H(X) &= -(1-P_e)\log(1-P_e) - (m-1) \times \frac{P_e}{m-1} \log \frac{P_e}{m-1} \\ &= -(1-P_e)\log(1-P_e) - P_e \log P_e + P_e \log(m-1) \\ &= H(P_e) + P_e \log(m-1). \end{aligned}$$

Entropy and Probability of Errors

Lemma

Consider two independent RVs defined over the same set: $X \sim p(x), X' \sim r(x')$. Then,

$$\Pr(X = X') \geq \max\left\{2^{-H(p,r)}, 2^{-H(r,p)}\right\}.$$

Proof: $2^{-H(p,r)} = 2^{\mathrm{E}_p(\log_2 r(X))} \leq \mathrm{E}_p\left(2^{\log_2 r(X)}\right) = \sum_{x \in \mathcal{X}} p(x)r(x) = \Pr(X = X')$.

Corollary

Let X, X' are i.i.d. with entropy $H(X)$. Then,

$$\Pr(X = X') \geq 2^{-H(X)},$$

with equality iff X has a uniform distribution.